Conceptual Framework

Page • Tag

3. Conceptual Framework

3.1 Strategic Behavior vs. Belief Updating

Much of the literature on misinformation and polarization models belief change as an informational process. Individuals are exposed to content, update beliefs through Bayesian or boundedly rational updating, and collective outcomes emerge through opinion diffusion. In these models, dynamics are primarily epistemic: agents change what they believe based on what they see.

Our framework takes a different starting point.

We are not modeling belief formation as such. We are modeling **strategic behavior under social reinforcement and institutional constraint**. The key object of interest is not whether an individual privately believes a piece of content, but whether they choose to *publicly propagate* it.

This distinction matters.

Belief updating is about internal cognitive states. Strategy adoption in our framework is about observable behavioral choices that spread when they generate reward. Sharing harmful or high-intensity content is treated as a strategy whose prevalence evolves based on its perceived payoff relative to alternatives.

Thus:

- Belief models explain *what people think*.
- Strategic adaptation models explain *what people do*.

Our model belongs to the second category.

We assume that individuals observe the social environment and adapt their behavior based on relative success. This adaptation occurs not through individual learning or belief revision, but through **population-level imitation**: strategies that generate attention, validation, or emotional reward increase in prevalence as more individuals adopt them. Strategies that incur costs decline in prevalence as individuals switch away.

The dynamics are therefore evolutionary and incentive-driven rather than purely informational. Strategy prevalence changes through imitative switching based on payoff differences, analogous to replicator dynamics in evolutionary game theory.

This shift reframes the problem of misinformation from epistemic distortion to incentive distortion.

3.2 Moderation as Social Governance

Online moderation is often treated as a technical filtering mechanism: remove harmful content, reduce exposure, improve information quality. In practice, however, moderation functions as a form of governance.

Governance has two defining features:

1. It imposes costs on certain behaviors.
2. Its legitimacy shapes how those costs are interpreted.

When a platform increases enforcement intensity—through bans, labeling, deboosting, or removal—it alters the payoff structure of user behavior. But unlike a purely mechanical penalty, governance interventions are interpreted through social meaning.

The same action can be perceived as:

- Legitimate enforcement of norms, or
- Hostile suppression of identity or voice.

This interpretive layer introduces a structural asymmetry: the deterrent effect of enforcement depends on institutional trust.

If trust is high, governance penalties reinforce norms and reduce harmful behavior. If trust is low, the same penalties may be reinterpreted as unjust or adversarial, reducing deterrent power without necessarily increasing the symbolic appeal of the targeted behavior.

Thus moderation is not merely an external constraint; it is a relational intervention embedded in a legitimacy context.

This creates the possibility that increasing enforcement does not monotonically reduce harmful behavior. Instead, its effectiveness depends on the state of institutional trust, which itself may evolve in response to governance actions.

3.3 Incentive Structure: Baseline Appeal, Social Reinforcement, and Governance Cost

Strategies spread through imitation of behaviors that generate perceived rewards.

In digital environments, reward signals often take the form of visibility, validation, attention, and emotional resonance. High-intensity content—especially content framed around outrage, threat, or moral violation—tends to generate stronger engagement and emotional response.

These signals create strategic incentives. We conceptualize the perceived payoff of harmful content sharing as having three structural components:

3.3.1 Baseline Emotional Appeal

Independent of prevalence, harmful or high-intensity content may carry intrinsic emotional reward. This includes:

- Satisfaction from expressing moral outrage
- Emotional release from sharing threatening narratives
- Identity affirmation through alignment with group values
- Psychological reward from challenging perceived authority

This baseline component represents a constant incentive present even when the strategy has low prevalence. It captures the "pull" of the content itself, absent social dynamics.

We denote this baseline appeal as **θ**.

3.3.2 Social Reinforcement

As a strategy becomes more prevalent, it generates additional reward through social validation and coordination. This includes:

- **Coordination benefits**: aligning with others reduces isolation and increases group cohesion
- **Social proof**: widespread adoption signals that the behavior is acceptable, important, or correct
- **Emotional contagion**: observing others express intensity increases one's own emotional arousal
- **Network effects**: the value of participation increases with the size of the participating community

We model this as **R(x)**, where x is the fraction of the population adopting the harmful strategy.

Baseline specification: $R(x) = \alpha x$ (linear reinforcement)

This represents contexts with strong coordination incentives, informational cascades, or identity dynamics where reinforcement increases monotonically with prevalence throughout the relevant range.

Alternative specifications tested for robustness:

- Saturating: $R(x) = \alpha x/(1+x)$
- Inverted-U: $R(x) = \alpha x(1-x)$
- Threshold: $R(x) = \alpha[1-\exp(-\lambda x)]$

The linear form is not a strong claim about reinforcement psychology, but rather a baseline that isolates the trust-mediated mechanism without additional nonlinearities. Robustness checks (Section 5.3) confirm that the core inversion phenomenon survives across all tested specifications.

3.3.3 Governance Cost, Mediated by Trust

Governance interventions impose costs on harmful behavior through removal, sanctions, deboosting, or social stigma. However, these costs are not perceived uniformly. Their deterrent effect depends on **institutional trust**.

We model the effective governance cost as **$g \cdot \Phi(T)$**, where:

- **g** is the intensity of enforcement (moderation effort, penalty severity, enforcement frequency)
- **$\Phi(T)$** is a trust-mediated effectiveness function
- **T** is the level of institutional trust

Effectiveness Function Specification

The function $\Phi(T)$ captures how institutional trust shapes the translation of enforcement intensity into perceived deterrent cost. We specify:

$\Phi(T) = T^\kappa$, where $\kappa > 1$

This functional form has several key properties:

1. Bounded and Positive

- $\Phi(T) \in (0, 1]$ for $T \in (0, 1]$
- Governance never literally increases the appeal of harmful behavior
- At zero trust, effectiveness vanishes but does not reverse sign

2. Strictly Increasing

- $\Phi'(T) = \kappa T^{(\kappa-1)} > 0$ for all $T > 0$
- Higher trust always translates enforcement into greater deterrent power

3. Convex (Accelerating Returns to Trust)

- $\Phi''(T) = \kappa(\kappa-1)T^{(\kappa-2)} > 0$ for $\kappa > 1$
- Marginal effectiveness of trust increases with trust level
- Moving from $T = 0.2$ to $T = 0.4$ has less impact than moving from $T = 0.6$ to $T = 0.8$

4. Curvature Parameter

- κ determines how sharply effectiveness declines in low-trust regimes
- $\kappa = 1$: Linear (no inversion possible)

- $\kappa = 2$: Quadratic (moderate nonlinearity)
- $\kappa = 3$: Cubic (strong trust-dependence)

Behavioral Interpretation

The power-law form captures the following mechanism:

High Trust ($T \approx 1$): Governance penalties are perceived as legitimate enforcement of shared norms. Sanctions carry full deterrent weight. Individuals interpret costs as:

- "This violates community standards"
- "The platform is protecting users"
- "I could face real consequences"

Medium Trust ($T \approx 0.5$): Governance effectiveness is substantially degraded. The same penalty is reinterpreted as:

- "Enforcement is inconsistent"
- "Others are getting away with it"
- "This feels arbitrary"

Deterrent power is present but weakened.

Low Trust ($T \approx 0.2$): Governance penalties are dismissed, ignored, or worked around. Not because they actively increase appeal, but because they fail to register as meaningful costs:

- "This doesn't apply to people like me"
- "They have no legitimate authority"
- "Everyone knows these rules aren't real"

The convexity ($\kappa > 1$) reflects a critical insight: **trust erosion has compounding effects on governance effectiveness**. Each unit decline in trust removes more deterrent power than the previous unit. This is consistent with research on institutional legitimacy showing that once trust falls below moderate levels, enforcement often generates minimal compliance.

Importantly, this formulation does **not** require symbolic resistance or reactance effects. It is a conservative representation: low trust means governance becomes ineffective, not counterproductive. Inversion emerges purely from the geometric relationship between enforcement intensity and trust-mediated effectiveness.

Analytical Derivation: The Comparative Elasticity Condition

We now derive the precise structural condition under which increasing governance intensity reduces effective deterrent cost.

Setup:

- Effective governance cost: $C(g,T) = g \cdot \Phi(T)$
- Trust dynamics at equilibrium (derived in Section 3.4): $T^*(g) = T_0 - (\beta/\gamma)g$
- Effective cost function: $F(g) \equiv g \cdot \Phi(T^*(g))$

At behavioral equilibrium:

```
θ + αx* = F(g)
x*(g) = [F(g) - θ]/α
```

Harmful behavior increases with enforcement when:

```
dx*/dg = F'(g)/α > 0  ⟺  F'(g) > 0
```

Computing the derivative for general trust erosion h(g):

Let $T(g) = T_0 - h(g)$, where $h(0) = 0$, $h'(g) > 0$ (monotonic erosion).

```
F(g) = g·Φ(T(g))

F'(g) = Φ(T) + g·Φ'(T)·T'(g)
      = Φ(T) - g·Φ'(T)·h'(g)
```

Inversion condition ($F'(g) < 0$):

```
g·Φ'(T)·h'(g) > Φ(T)
```

Rearranging:

```
g·h'(g) > Φ(T)/Φ'(T)
```

Interpretation: Comparative Elasticities

The right-hand side, $\Phi(T)/\Phi'(T)$, measures the **elasticity of effectiveness** with respect to trust. For power-law effectiveness $\Phi(T) = T^\kappa$:

```
Φ(T)/Φ'(T) = T/κ
```

The left-hand side, $g \cdot h'(g)$, measures the **local intensity of trust erosion**: how much additional trust is lost per unit increase in enforcement, scaled by current enforcement level.

The inversion condition states:

> **Trust erosion intensity must exceed effectiveness elasticity.**

Formally:

```
Trust erosion intensity     >  Effectiveness sensitivity
      g·h'(g)               >      T/κ
```

This is a **comparative rates result**: inversion occurs when the rate at which enforcement erodes institutional capacity exceeds the rate at which it mobilizes resources.

Baseline Case: Linear Trust Erosion

For $h(g) = cg$ (linear erosion), we have $h'(g) = c$ (constant).

Inversion condition:

```
g·c > T/κ
```

Substituting $T^*(g) = T_0 - cg$:

```
gc > (T₀ - cg)/κ
κgc > T₀ - cg
```

```
gc(κ + 1) > T₀
```

Critical enforcement threshold:

```Plain Text
g* = T₀/[c(κ + 1)] = T₀γ/[β(κ + 1)]
```

Three enforcement regimes:

Low enforcement (g < g):*

- Trust remains relatively high
- Effectiveness gain from additional enforcement outweighs trust erosion
- F'(g) > 0: **Normal regime**
- Higher enforcement → higher effective cost → lower harmful behavior

Optimal enforcement (g = g):*

- F(g) reaches maximum
- Marginal benefit of enforcement exactly balances marginal cost of erosion
- Peak deterrence capacity

High enforcement (g > g):*

- Trust has declined substantially
- Effectiveness loss from erosion outweighs intensity gain
- F'(g) < 0: **Inversion regime**
- Higher enforcement → lower effective cost → higher harmful behavior

---

Numerical Example

Parameters:

```Plain Text
T₀ = 0.7     (baseline trust)
β = 0.15     (erosion rate, adjusted for broader range)
γ = 0.3      (restoration rate)
c = β/γ = 0.5
κ = 2        (quadratic effectiveness)
θ = 0.02     (baseline appeal)
α = 0.08     (reinforcement coefficient)
```

Critical threshold:

```
g* = 0.7/[0.5(1 + 2)] = 0.467
```

Trust constraint: $T^*(g) \geq 0$ requires $g \leq T_o/c = 1.4$

At g = 0.3 (below optimal):

```
T*(0.3) = 0.7 - 0.5(0.3) = 0.55
F(0.3) = 0.3 × (0.55)² = 0.091
x*(0.3) = [0.091 - 0.02]/0.08 = 0.89
```

At g = 0.467 (optimal):

```
T*(0.467) = 0.7 - 0.5(0.467) = 0.467
F(0.467) = 0.467 × (0.467)² = 0.102
x*(0.467) = [0.102 - 0.02]/0.08 = 1.03 → bounded at 1.0
```

At g = 0.8 (above optimal):

```
T*(0.8) = 0.7 - 0.5(0.8) = 0.30
F(0.8) = 0.8 × (0.30)² = 0.072
x*(0.8) = [0.072 - 0.02]/0.08 = 0.65
```

Result: Moving from below-optimal (g = 0.3) to above-optimal (g = 0.8):

- Enforcement intensity increases by 167%
- Effective deterrent cost **decreases** by 21%
- Harmful behavior prevalence **decreases** from 0.89 to 0.65

Note: In this calibration, the system operates near the boundary (x close to 1), reflecting the marginal deterrence regime where governance is struggling but not completely ineffective. The inversion is manifest in F(g), though behavioral responses may be truncated by boundary effects.

Robustness to Alternative Erosion Specifications

Convex erosion: $h(g) = cg^v$, $v > 1$

Erosion intensity: $g \cdot h'(g) = g \cdot cvg^{(v-1)} = cvg^v$

This **increases with g**, making the inversion inequality easier to satisfy at high enforcement levels.

Critical threshold shifts to:

```
Plain Text

g* = [T₀/(c(κv + 1))]^(1/v)
```

Effect: Convex erosion produces:

- Later onset of inversion (g^* higher)
- Sharper decline in $F(g)$ past the peak
- Potentially narrower inversion window if trust boundary binds

Sublinear erosion: $h(g) = c\sqrt{g}$

Erosion intensity: $g \cdot h'(g) = c\sqrt{g}/2$

This **increases slower than linear**, making inversion less likely.

General principle: The shape of $h(g)$ determines how erosion intensity $g \cdot h'(g)$ scales with enforcement. Faster-than-linear erosion ($v > 1$) widens the parameter range supporting inversion; slower-than-linear erosion ($v < 1$) narrows it.

Alternative Effectiveness Specifications

Sigmoid:

```
Plain Text

Φ(T) = 1/(1 + e^(-μ(T - T₀)))
```

Sharp threshold effect around $T = T_0$. Inversion occurs when trust crosses the inflection point where $\Phi'(T)$ is largest.

Affine + Power:

```
Plain Text

Φ(T) = ε + (1-ε)T^κ
```

Governance retains minimum effectiveness ε even at zero trust (floor effect). Inversion persists as long as ε is small (< 0.2).

Result: All strictly increasing, sufficiently convex specifications produce inversion when the comparative elasticity condition is met. The power-law form $\Phi(T) = T^\kappa$ is used for analytical tractability, not because it is uniquely capable of generating the effect.

What Is NOT Required

The inversion mechanism does **not** require:

❌ Governance to become counterproductive ($\Phi(T) < 0$)
❌ Symbolic reactance or psychological rebellion
❌ Adversarial coordination or strategic evasion
❌ Algorithmic amplification or platform recommendation
❌ Network structure or preferential attachment
❌ Agent heterogeneity in trust or susceptibility

It requires only:

✓ Monotonic trust erosion from enforcement visibility
✓ Convex effectiveness function (accelerating returns to trust)
✓ Comparative elasticity condition: $g \cdot h'(g) > \Phi(T)/\Phi'(T)$

3.3.4 Combined Payoff Structure

The perceived payoff difference between harmful sharing and safe alternatives is:

$$\Delta\pi = \theta + \alpha x - g \cdot \Phi(T)$$

This formulation makes clear that:

- **Incentives are endogenous**: they depend on the behavioral state (x) and institutional state (T)
- **Governance effectiveness is contingent**: the same enforcement intensity g has different effects depending on trust
- **Positive feedback is structural**: social reinforcement creates inherent amplification independent of governance

When trust is high, governance cost dominates and harmful behavior is deterred. When trust erodes sufficiently, reinforcement and baseline appeal can overcome governance cost, potentially reversing the strategic ranking.

3.4 Trust as a State Variable

Institutional trust is not a fixed parameter. It evolves in response to governance actions and social context.

Trust is treated as a **state variable**, meaning:

- It changes over time
- It depends on past governance intensity
- It influences future behavioral incentives through cost mediation

This introduces **memory** into the system. Past enforcement affects current legitimacy, which alters the present interpretation of governance costs.

This is structurally crucial.

If trust were constant, governance would have stable meaning. Increasing enforcement would always increase deterrent costs in a predictable way, and the relationship between enforcement intensity and behavioral outcomes would be monotonic.

But if trust evolves in response to governance, then the meaning of enforcement becomes **endogenous**. Governance changes trust; trust changes the effectiveness of governance. This feedback creates the possibility of **time-dependent payoff rankings**.

A strategy that is unattractive under high trust may become attractive under low trust—not because its intrinsic reward changed, but because the perceived cost changed through legitimacy mediation.

Trust Dynamics: Erosion and Restoration

In our formalization, trust responds to **governance visibility** rather than effectiveness per se. The dynamics are:

$$\dot{T} = -\beta g + \gamma(T_0 - T)$$

where:

- **$-\beta g$**: Trust erosion from enforcement visibility. Sustained or high-intensity governance creates resentment, perceptions of overreach, or loss of procedural fairness—regardless of whether enforcement is technically effective in reducing harmful behavior in the short term.
- **$\gamma(T_0 - T)$**: Restoration toward baseline trust T_0. In the absence of active enforcement pressure, legitimacy gradually recovers as negative associations fade and baseline institutional attitudes reassert themselves.

At equilibrium ($\dot{T} = 0$):

```
Plain Text

βg = γ(T₀ - T*)
T* = T₀ - (β/γ)g
```

This specification reflects an important empirical regularity: sustained high-visibility enforcement can erode legitimacy even when effective, because the *perception of being governed intensively* itself generates resistance in contested contexts.

Trust restoration occurs slowly (small γ) relative to behavioral adaptation, consistent with findings that institutional legitimacy is sticky and recovers gradually after crises.

Scope Condition: Contested Governance Contexts

This trust erosion mechanism is most applicable in contexts where:

- **Governance is socially contested**: Different groups hold divergent views on the legitimacy of enforcement
- **Enforcement is visible**: Users are aware of and experience moderation actions
- **Institutional authority is questioned**: The platform/institution does not have universal legitimacy
- **Identity stakes are high**: Enforcement touches on values, beliefs, or group membership

It is less applicable when:

- Enforcement is covert or invisible
- Legitimacy is strongly institutionalized and stable
- The governed population has homogeneous normative expectations
- Penalties are purely material (fines, jail) rather than socially interpreted

This is not a limitation but a scope specification. Many real-world governance challenges—including content moderation, community norm enforcement, and regulatory compliance in polarized contexts—operate precisely in these contested institutional environments where trust becomes decisive.

Extensions: Alternative Trust Dynamics

Effectiveness-contingent erosion:

```Plain Text
Ṫ = -β·g·[1 + η(x - x_target)] + γ(T₀ - T)
```

Trust erodes faster when enforcement is visible *and failing* (high x despite high g). This captures the intuition that ineffective governance damages legitimacy more than effective governance.

Behavioral restoration:

```
Ṫ = -βg + γ(T₀ - T)·(1 - x/x_threshold)
```

Trust recovers faster when harmful behavior is low, reflecting validation of governance legitimacy through perceived success.

Nonlinear erosion:

```
Ṫ = -β·g^ν + γ(T₀ - T),  ν > 1
```

Erosion accelerates at high enforcement (overreach effect becomes pronounced).

The baseline linear specification $T^*(g) = T_0 - (\beta/\gamma)g$ is chosen for analytical tractability and provides a conservative baseline. Future work can systematically explore these alternatives, though preliminary numerical analysis suggests the core inversion mechanism persists across specifications.

3.5 Scope Condition: The Marginal Deterrence Regime

The inversion mechanism operates in a specific institutional regime characterized by **marginal deterrence**: contexts where governance is strong enough to influence behavior but not strong enough to eliminate the targeted strategy entirely.

Formally, this requires:

```
Peak governance effectiveness ≈ Total reward
F_max ≡ F(g*) ~ θ + α
```

Three deterrence regimes:

Overwhelming Deterrence (F_max >> θ + α):

- Even minimal enforcement eliminates harmful behavior
- $x^* \to 0$ for all feasible g
- Inversion irrelevant (problem already solved)
- Examples: Severe legal penalties, strong cultural taboos, contexts with very high baseline trust

Marginal Deterrence (F_max ~ θ + α):

- Governance sufficient to suppress but not eliminate
- Interior equilibrium maintained across g range
- **Inversion mechanism active**
- Examples: Content moderation, norm enforcement, tax compliance, contested regulations

Ineffective Deterrence ($F_{max} << \theta$):

- Governance too weak to deter at any intensity
- $x^* \to 1$ for all feasible g
- Inversion irrelevant (governance already failed)
- Examples: Weakly enforced regulations, collapsed institutional authority, low-capacity states

Empirical implication: The trust-mediated inversion mechanism is most relevant in contexts where:

- Enforcement is visible and contested (not covert or universally accepted)
- Penalties are moderate (not trivial, not catastrophic)
- Social reinforcement provides significant counter-pressure to governance
- Institutional legitimacy is variable and responsive (not permanently high or collapsed)
- Baseline appeal and social rewards are of comparable magnitude to governance capacity

This scope condition is not a limitation but rather a specification of the domain of applicability. Many real-world governance challenges operate precisely in this marginal deterrence regime where institutional trust becomes decisive.

3.6 Why Failure Is Structural, Not Accidental

The framework does not claim that moderation always fails. Nor does it claim that backlash is inevitable.

Instead, it isolates conditions under which policy failure becomes **structurally possible**—a consequence of system dynamics rather than implementation error or adversarial response.

Failure in this context means:

An increase in governance intensity leads to an increase in the prevalence of the targeted harmful strategy, or equivalently, a decrease in effective deterrent capacity.

This is not attributed to:

- Platform incompetence or poor policy design
- Random shocks or exogenous events
- Adversarial manipulation or coordinated evasion
- Algorithmic amplification artifacts
- Network cascade effects

Rather, it emerges from the **coupled dynamics** of:

1. **Imitative strategy adoption** based on relative payoffs
2. **Social reinforcement** of prevalent behaviors
3. **Trust-mediated reinterpretation** of governance costs
4. **Governance-induced trust erosion**

The Mechanism: Comparative Rates of Mobilization vs. Erosion

Traditional deterrence theory assumes:

```
Deterrence Capacity = Enforcement Intensity
```

Trust-mediated deterrence recognizes:

```
Deterrence Capacity = g × Φ(T(g))
                    = Enforcement × Legitimacy(Enforcement)
```

Critical insight: Enforcement has **dual effects**:

1. **Direct effect:** More resources (↑g) increase capacity
2. **Indirect effect:** Enforcement visibility erodes legitimacy (↓T), reducing the effectiveness of those resources

Net effect depends on comparative rates:

```
Marginal benefit:  Φ(T)           (effectiveness at current trust)
Marginal cost:     g·h'(g)·Φ'(T)  (erosion intensity × sensitivity)
```

Inversion occurs when:

```
g·h'(g)·Φ'(T) > Φ(T)
```

Interpretation:

> When the rate at which enforcement erodes institutional capacity exceeds the rate at which it mobilizes deterrent resources, effective deterrence declines with enforcement intensity.

This is a comparative elasticity condition, not a behavioral paradox.

It reflects a fundamental institutional trade-off: resource mobilization versus capacity erosion. When erosion outpaces mobilization, more enforcement produces less deterrence.

Generalization Beyond Content Moderation

This mechanism applies to any institutional context where:

- **Capacity = Resources × Legitimacy**
- **Legitimacy declines with resource mobilization** (enforcement visibility)
- **Effectiveness has convex returns in legitimacy** (trust amplifies impact)

Examples:

Policing in contested neighborhoods:

- More patrols (↑g) increase visibility
- But erode community trust (↓T)
- If trust loss outpaces patrol increase, cooperation declines
- Crime deterrence can decrease with enforcement intensity

Tax enforcement in polarized contexts:

- More audits (↑g) increase detection
- But signal institutional hostility (↓T)
- If perceived legitimacy collapses, compliance becomes symbolic resistance
- Tax evasion can increase with audit intensity

Regulatory compliance under institutional distrust:

- Stricter inspections (↑g) improve monitoring
- But erode industry cooperation (↓T)
- If firms view regulator as adversarial, they minimize formal compliance

- Substantive violations can increase with regulatory intensity

Community norm enforcement:

- More active moderation (↑g) removes violations
- But signals community conflict (↓T)
- If members view norms as imposed, not shared, enforcement backfires
- Norm violations can increase with moderation intensity

The contribution: Showing that this institutional trade-off produces non-monotonic policy effects through geometric structure alone, without requiring strategic evasion, network amplification, or psychological rebellion.

3.7 Modeling Strategy: Minimal vs. Full Dynamics

The analysis proceeds in two stages to isolate causal mechanisms:

Stage 1: Minimal Model (2D) — Controlled Isolation

State variables: x(t), T(t)

Parameters: g (governance intensity, exogenous)

Purpose: Isolate the pure trust-mediated inversion mechanism

Research question: Can trust erosion alone cause governance to reduce its own effectiveness, absent any feedback dynamics?

What it deliberately abstracts from:

- Reactive governance (enforcement responding to behavior)
- Agent heterogeneity (population variation in trust/susceptibility)
- Network structure (spatial diffusion patterns)
- Temporal disequilibrium (dynamics away from steady state)
- Strategic adversaries (coordinated evasion)

Why this abstraction is valuable:

Scientific isolation: By treating g as exogenous, we ask: "Suppose a policymaker sets enforcement at level g. What equilibrium (x*, T*) emerges?" This comparative static cleanly isolates the trust mechanism from feedback dynamics.

Conservative baseline: If inversion occurs even without governance feedback, network effects, or adversarial response, then adding these factors can only amplify the phenomenon (or leave it unchanged). The minimal model provides a lower bound on the severity of the problem.

Conceptual clarity: Separating the trust-mediated effectiveness loss (2D mechanism) from governance-behavior feedback loops (3D dynamics) allows

precise attribution of effects.

Empirical interpretation: Real systems exhibit both mechanisms simultaneously. Observational data cannot cleanly separate them. The 2D model tells us what to expect from pure trust erosion; deviations indicate the importance of feedback dynamics.

Positioning for reviewers: We are not claiming real platforms operate at fixed g. We are claiming that *if they did*, trust erosion would produce non-monotonic effectiveness. This proves the mechanism exists independently of policy adaptation.

Stage 2: Full Model (3D) — Dynamic Consequences

State variables: x(t), T(t), G(t)

All variables endogenous

Governance response:

```
Ġ = α(x - x_target)
```

Purpose: Show what happens when policy adapts to behavior in a system with trust-mediated effectiveness

Additional dynamics:

- **Runaway collapse:** x ↑ → G ↑ → T ↓ → effectiveness ↓ → x ↑ [self-reinforcing]
- **Overshooting:** Governance responds to current x, but trust adjusts slowly; by the time G peaks, T is too low for it to work
- **Bistability:** Multiple equilibria (low-enforcement/low-harm vs. high-enforcement/high-harm)
- **Path-dependence:** Small shocks can push from good to bad equilibrium
- **Hysteresis:** Once in collapse regime, reducing G may not restore low-harm state

The 3D model does not change the core insight (trust erosion can make governance lose effectiveness). It shows what happens when policy chases a target in a system with this property.

Result: Feedback can amplify the inversion problem, creating sustained policy failure even when the underlying mechanism would produce only temporary non-monotonicity under fixed enforcement.

3.8 Falsifiability and Empirical Implications

Theoretical Falsification

The mechanism fails if:

1. **Inversion never occurs** under empirically plausible parameter ranges ($\theta \in [0.01, 0.1]$, $\kappa \in [1.5, 3.0]$, $\beta/\gamma \in [0.3, 2.0]$)
2. **Inversion only appears** under knife-edge parameter combinations requiring extreme nonlinearities with no behavioral justification
3. **Removing trust-mediation** (fixing T) does not restore monotonic governance effects
4. **The comparative elasticity inequality** $g \cdot h'(g) > \Phi(T)/\Phi'(T)$ has no empirically observable correlates

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Empirical Predictions

Prediction 1: Non-Monotonic Enforcement Effects

In contexts with measurable trust and enforcement intensity, the relationship between g and harmful behavior prevalence x should exhibit non-monotonicity when:

- Baseline trust T_0 is moderate (not extremely high or low)
- Trust is responsive to enforcement (β/γ substantive)
- Effectiveness curvature κ is sufficient ($\gtrsim 1.5$)

Testable implication: Plotting x against g (controlling for confounds) should reveal an inverted-U or hump-shaped relationship, with a critical threshold g* where marginal effects reverse.

Prediction 2: Trust-Mediated Heterogeneity

Contexts with higher baseline trust T_0 should exhibit:

- Higher optimal enforcement threshold g*
- Wider inversion window
- Smaller magnitude of backfire effect

Testable implication: Cross-sectional variation in trust should predict variation in the enforcement intensity at which marginal effects reverse.

Prediction 3: Elasticity-Based Threshold

The critical enforcement level g* should satisfy:

```
g⋆·h'(g⋆) ≈ T⋆(g⋆)/κ
```

Testable implication: Estimating h(g) and Φ(T) separately allows prediction of g* without observing the full enforcement-behavior relationship.

Estimation strategy:

Step 1: Estimate trust response to enforcement

```
Regress T on g (and controls) → recover h(g)
Compute h'(g) numerically
```

Step 2: Estimate effectiveness response to trust

```
Regress compliance/violation on T (and controls) → recover Φ(T)
Compute Φ'(T) numerically
```

Step 3: Predict inversion threshold

```
Solve: g·h'(g)·Φ'(T) = Φ(T) for g
Compare to observed g⋆ from reduced-form analysis
```

Prediction 4: Differential Effects by Governance Regime

Systems in the marginal deterrence regime ($F_{max} \sim \theta + \alpha$) should exhibit inversion, while systems in overwhelming deterrence ($F_{max} >> \theta + \alpha$) or ineffective deterrence ($F_{max} << \theta$) should show monotonic relationships.

Testable implication: Identifying the deterrence regime ex ante (via baseline measures of governance capacity and behavioral incentives) should predict whether inversion is observable.

Prediction 5: Temporal Dynamics Under Adaptive Governance

In the 3D model with endogenous enforcement ($\dot{G} = \alpha(x - x_{target})$), systems should exhibit:

- Possible runaway escalation (G and x both increasing)
- Hysteresis (path-dependence in response to shocks)
- Difficulty in stabilizing at optimal enforcement g*

Testable implication: Time-series data should show:

- Positive correlation between enforcement growth and harmful behavior growth in some periods (inversion regime)
- Divergence from target x_target despite increasing G
- Asymmetric response to enforcement increases vs. decreases

Boundary Conditions for Empirical Testing

The mechanism is most empirically relevant when:

✓ Trust is measurable (surveys, revealed preference, proxy measures)

✓ Enforcement intensity is observable (moderation actions, audit rates, patrol density)

✓ Behavioral outcomes are quantifiable (violation rates, compliance measures)

✓ Governance is contested (not universally accepted)

✓ Time-series or cross-sectional variation in g exists

✓ Confounds can be controlled (selection, composition, external shocks)

It is less testable when:

× Enforcement is covert or unmeasured

× Trust is fixed or unobservable

× Behavioral data is unavailable

× Variation in enforcement is endogenous to unobserved factors

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3.9 Summary: The Core Institutional Mechanism

Traditional deterrence theory:

```
Deterrence = Enforcement Intensity
```

Trust-mediated deterrence:

```
Deterrence Capacity = g × Φ(T(g))
                    = Enforcement × Legitimacy(Enforcement)
```

Key structural insight: Enforcement has dual effects:

1. Direct effect: ↑g increases resources
2. Indirect effect: ↑g erodes legitimacy, reducing effectiveness of those resources

Inversion occurs when:

```
Rate of capacity erosion > Rate of resource mobilization
      g·h'(g)·Φ'(T)      >      Φ(T)
```

This is a comparative elasticity condition: the rate at which enforcement erodes trust (weighted by enforcement level) must exceed the rate at which trust amplifies effectiveness.

Not required:

- Symbolic rebellion ($\Phi(T) > 0$ throughout)
- Adversarial coordination
- Algorithmic amplification
- Network structure
- Agent heterogeneity

Required:

- Monotonic trust erosion from enforcement visibility
- Convex effectiveness ($\kappa > 1$)
- Sufficient erosion intensity (β/γ not too small)
- Marginal deterrence regime ($F_{max} \sim \theta + \alpha$)

Generalization: This mechanism applies to any institutional context where capacity equals resources times legitimacy, and legitimacy declines with resource mobilization. Examples include policing, tax enforcement, regulatory compliance, and community norm enforcement—all contexts where institutional trust mediates enforcement effectiveness.

Contribution: Demonstrating that non-monotonic policy effects can emerge from pure institutional-geometric relationships, without behavioral paradoxes or strategic

responses. The mechanism is structural, testable, and generalizable beyond content moderation.

Phase 1 establishes: The trust-mediated inversion mechanism exists and can be analytically characterized.

Phase 2 will examine: Dynamic consequences under adaptive governance, empirical calibration, and agent-based heterogeneity.

END OF CONCEPTUAL FRAMEWORK